



Yacht Hydrodynamic Resistance

Ata Atasoy Peng Rao
Hüseyin Boran Kuşcu Gonzalo Bengoa Díaz

June 2026

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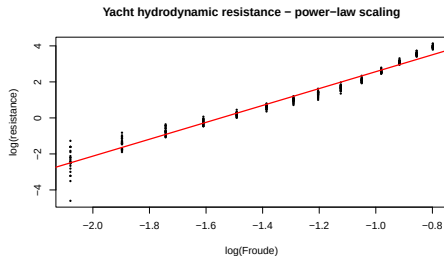
Part 1

Data and Problem

The Yacht Hydrodynamics dataset

Source. UCI ML Repository: Delft Systematic Yacht Hull Series, towing-tank experiments.

- 22 hull forms, each tested at several speeds
- $n = 308$ observations (one hull $\times$ one speed per row)
- **Response:** residuary resistance per unit weight of displacement
- **Predictors:** speed (Froude number) + five hull form factors, constant within a hull



- The near-linear pattern on log–log axes supports a power-law relation.
- Low- Fn scatter motivates checking residual variation in later models.

Part 2

Log–log Regression

Model and reference prior

Goal. Verify the physical power law resistance $\propto Fn^{\gamma_F}$ with $\gamma_F \approx 4$. Taking logs makes the exponent a regression slope:

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(Fn) + \sum_{k=1}^5 \beta_k \text{hull}_k + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

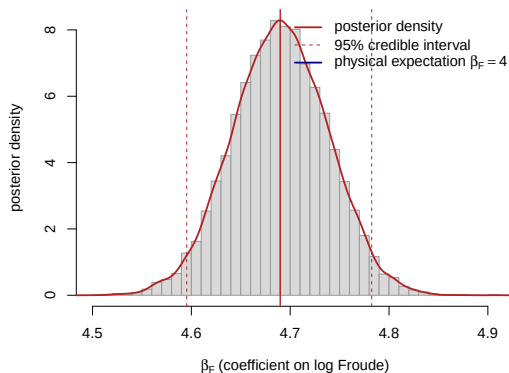
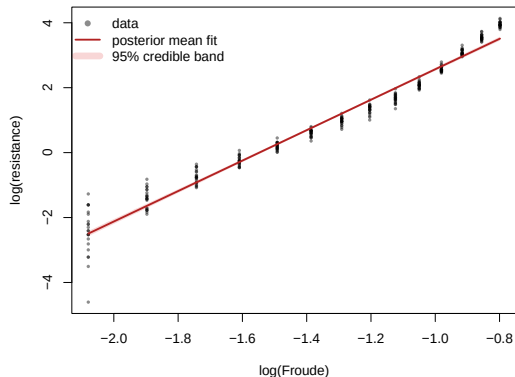
Baseline non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \propto 1, \quad \pi(\sigma^2) \propto \sigma^{-2}.$$

Closed-form posterior

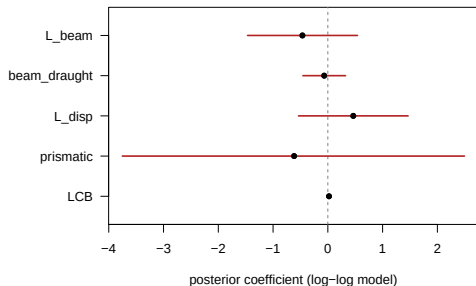
$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}(\hat{\boldsymbol{\beta}}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$; marginalising σ^2 gives a multivariate Student- t with $n - p$ d.o.f. The 95% credible intervals coincide with the OLS confidence intervals, no simulation required.

Result: the Froude exponent



- Excellent fit: $R^2 = 0.97$, residual SD $s = 0.32$ on the log scale.
- $\beta_F \approx 4.69$, 95% CI = $[4.60, 4.78]$, $\Pr(\beta_F > 4 \mid \mathbf{y}) \approx 1$.
- Power law confirmed, the exponent is *slightly steeper* than the textbook Fn^4 .

Hull covariates & residual diagnostics



Hull form factors. Once Fn is in the model, every 95% interval contains zero, the factors vary only across the 22 hulls and are strongly collinear (resolved by BMA in Part 3).

Residual diagnostics expose two defects:

- U-shaped trend vs. fitted values, a single constant exponent is too rigid;
- spread shrinks as the fitted value grows, textbook *heteroscedasticity*.

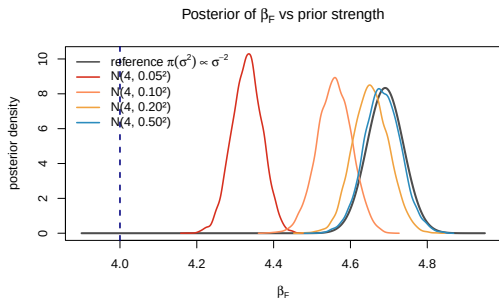
Both motivate the nonlinear correction in Part 4.

Prior sensitivity: does the prior drive $\beta_F > 4$?

Encode the physics in **independent informative priors** and ask how hard they must insist on 4:

$$\beta_F \sim \mathcal{N}(4, \tau^2), \quad \beta_{j \neq F} \sim \mathcal{N}(0, 100^2), \quad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}),$$

sampled with a **Gibbs sampler** (10^4 iters, 2×10^3 burn-in), sweeping τ .



Prior on β_F	Mean	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	1.00
reference prior	4.69	1.00

Only a near-dogmatic $\tau = 0.05$ drags the mean to 4.33. For any diffuse prior ($\tau \geq 0.1$) the data dominate: $\beta_F > 4$ is **robust to the prior**.

Part 3

BAS / BMA Model Selection

Model space: which hull factors matter after Froude?

Conditional question. Once the dominant physical predictor is fixed, which hull-geometry variables still carry signal?

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(Fn) + \sum_{k \in \mathcal{M}} \beta_k \text{hull}_k + \varepsilon.$$

- $\log(Fn)$ is included in every candidate model.
- Optional hull covariates: LCB, prismatic, L_disp, beam_draught, L_beam.
- Five optional variables imply $2^5 = 32$ submodels.
- BAS/BMA averages over models instead of relying on one saturated regression.

Question

Among the hull factors, which ones still matter conditional on Froude?

Baseline result: beam-draught has the largest PIP

Posterior inclusion probability (PIP) is the posterior probability that a covariate belongs in the regression model after averaging over all 32 submodels.

Covariate	PIP	Evidence
beam_draught	0.565	moderate
prismatic	0.265	weak
LCB	0.210	weak
L_disp	0.206	weak
L_beam	0.192	weak

- Highest posterior model: $\log(Fn) + \text{beam_draught}$.
- Its posterior probability is only 0.234, so model uncertainty remains substantial.
- beam_draught carries the strongest residual signal, but evidence is not decisive.

Takeaway. Froude is robust; hull effects are weaker, with beam-draught as the best secondary candidate.

Sensitivity: is the BAS/BMA ranking robust to the prior?

The baseline analysis was checked under alternative BAS/BMA prior specifications, with the same 32-model space and $\log(Fn)$ forced into every model.

Prior	LCB	prismatic	L_disp	beam_draught	L_beam
BIC / reference	0.210	0.265	0.206	0.565	0.192
g -prior ($g = n$)	0.186	0.217	0.182	0.499	0.166
JZS	0.060	0.057	0.051	0.243	0.042

- beam_draught is top-ranked under all three specifications.
- Its PIP falls from 0.565 to 0.243, so selection is prior-sensitive.
- No hull covariate receives consistently strong support.

Robust conclusion

Not “beam-draught is definitely selected”, but “beam-draught is the most defensible secondary hull covariate.”

From BAS/BMA to the nonlinear model

Log-log regression $\implies$ BAS/BMA over hull factors
 $\implies$ Nonlinear models M_0 vs. M_1 .

- BAS/BMA proposes `beam_draught` as the best-supported secondary hull covariate.
- Because evidence is moderate, we do not assume it is certainly important.
- Next step:

M_0 : Froude only, M_1 : Froude + `beam_draught`.

- The nonlinear heteroscedastic model checks whether this candidate improves prediction on the original resistance scale.

Takeaway

BAS/BMA is used as a screening step: it narrows the hull variables to the most defensible one, but leaves the final assessment to the nonlinear model comparison.

Part 4

Heteroscedasticity-Corrected Model

The model: Gamma likelihood with a learned variance

Back to the **original** (positive) resistance scale. We model $Y_i > 0$ with a **Gamma** likelihood, moment-matched to a mean μ_i and a variance σ_i^2 :

$$Y_i \sim \text{Gamma}(\alpha_i, r_i), \quad \alpha_i = \frac{\mu_i^2}{\sigma_i^2}, \quad r_i = \frac{\mu_i}{\sigma_i^2} \implies \mathbb{E}[Y_i] = \mu_i, \quad \text{Var}[Y_i] = \sigma_i^2.$$

Mean: the same power law (log link)

$$\mu_i = k \text{Fn}_i^{\gamma_F} \exp(\beta_B b d_i) = \exp(\mathbf{x}_i^\top \boldsymbol{\theta}).$$

Variance: let it grow

$$\sigma_i = \sigma_0 \mu_i^\delta.$$

δ is the heteroscedasticity correction

$\delta = 0$: constant variance | $\delta = 1$: multiplicative (constant CV — the log-log case). We *learn* δ instead of assuming it.

Two models carried over from BAS/BMA: M_0 (Froude only) vs. M_1 (Froude + beam_draught).

Priors: weakly informative, physics-anchored

Mean parameters

$$\log k \sim \mathcal{N}(0, 10^2), \quad \gamma_F \sim \mathcal{N}(4, 2^2), \quad \beta_B \sim \mathcal{N}(0, 2^2).$$

Variance-function parameters

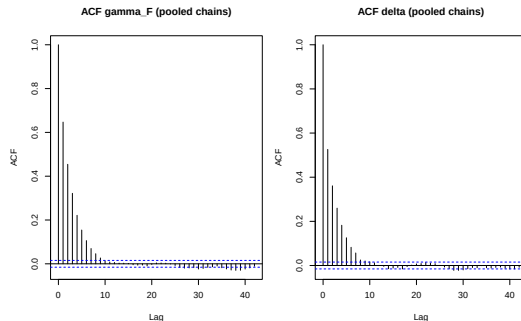
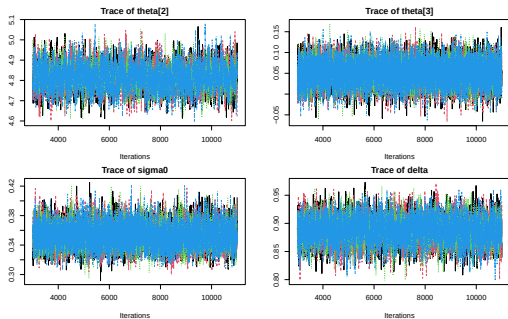
$$\log \sigma_0 \sim \mathcal{N}(0, 10^2), \quad \delta \sim \mathcal{N}(0, 1^2).$$

- γ_F centred on the **physical value** 4, but loose (sd = 2): the data, not the prior, drive the exponent.
- δ centred at 0 (**homoscedasticity**): a non-constant variance must be *earned* from the data.
- Remaining priors are vague but proper.

Prior sensitivity on γ_F

M_1 is re-fitted with a **vague** $\mathcal{N}(0, 10^2)$ and an **informative** $\mathcal{N}(4, 0.5^2)$ prior to confirm the exponent is governed by the likelihood, not the prior.

Inference: JAGS with slice sampling — M1 diagnostics



M1 parameter	b_0	γ_F	β_B	σ_0	δ
$\hat{R}$	1.0002	0.9999	1.0003	1.0000	1.0000
ESS	4409	2182	13012	3284	2208

Result 1: did learning the variance fix it?

Constant variance ($\delta = 0$) vs. learned variance (δ free) — same mean structure (M1)

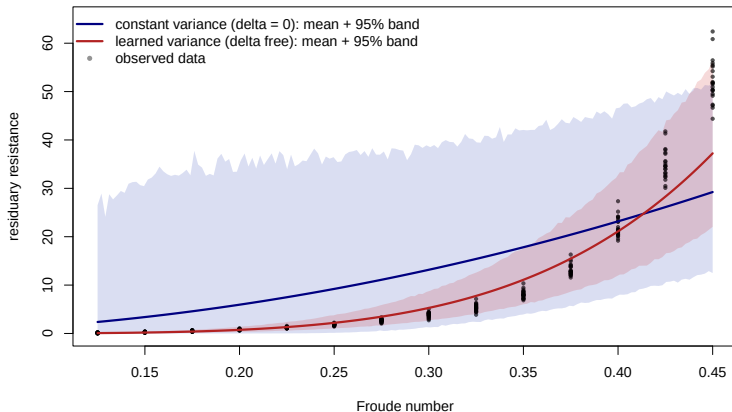
Model	WAIC	Δ WAIC	Δ elpd (se)	γ_F	cov.	width
Constant variance ($\delta = 0$)	1667.7	827.1	-413.5 (14.3)	1.97	0.968	35.8
Learned variance (δ free)	840.6	0.0	—	4.82	0.964	9.6

Variance growth

$\hat{\delta} \approx 0.89$, $\Pr(\delta > 0 \mid \mathbf{y}) = 1$. Δ WAIC = 827 (≈ 29 SE): constant variance is rejected, it *biases* the exponent to 1.97, and covers only with a uniform band $\approx 3.7\times$ wider.

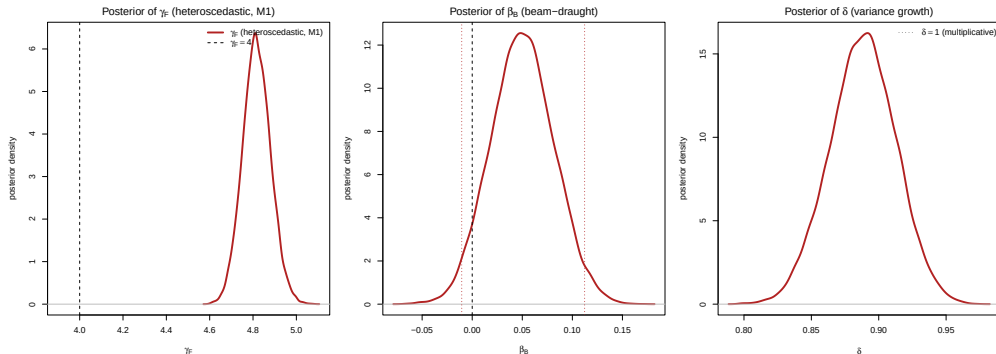
Result 1 (cont.): posterior predictive bands

Posterior predictive: constant vs learned variance (Task 3)



Same $\approx 96\%$ coverage, but constant variance needs a uniform tube (width 35.8), while the learned variance hugs the data (width 9.6).

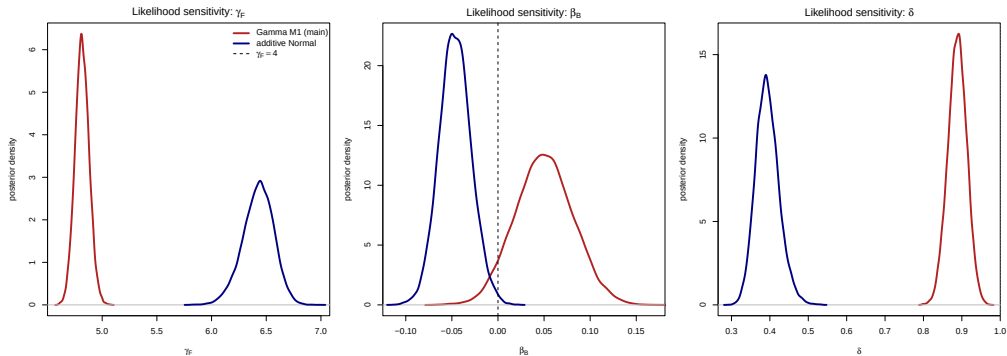
Result 2: the Froude exponent survives



Model	WAIC	LOOIC	p_{WAIC}	$\max \hat{k}$	$\Delta WAIC$	$\Delta \text{elpd (se)}$
M_0 : Froude only	840.9	841.0	5.8	0.6	0.3	-0.2 (1.9)
M_1 : + beam_draught	840.6	840.7	7.1	0.4	0.0	—

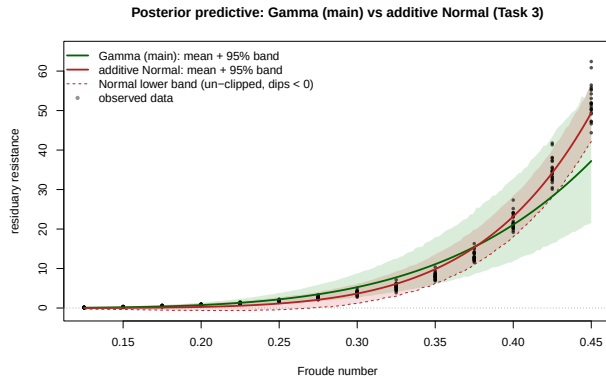
Sensitivity: the exponent is likelihood-dependent

Same mean, same variance function, only the **likelihood shape** changes (Gamma vs. additive Normal):



- Gamma keeps $\gamma_F \approx 4.8$; the Normal **inflates it to** ≈ 6.4 (squared-error weighting lets a few large points dominate).

Sensitivity: the exponent is likelihood-dependent



Conclusion

WAIC is a **tie** (840.6 vs. 842.8) — the data cannot separate them on fit, *but* the Normal puts $\approx 13\%$ of predictive mass **below zero** (impossible for resistance).

- Heteroscedasticity is captured: $\delta \approx 0.89$, $\Pr(\delta > 0 \mid \mathbf{y}) = 1$. Learning the variance is decisive ($\Delta\text{WAIC} \approx 827$), and necessary for the physics: constant variance biases the exponent to 1.97, vs. 4.82 when the variance is learned.
- The power law is robust: $\gamma_F \approx 4.82 \approx \beta_F \approx 4.69$, close to the textbook value 4.
- Hull effect does not survive: `beam_draught` is not credible under the positive-support model.
- The exponent is likelihood-sensitive (Normal $\rightarrow \approx 6.4$), so the error model must be reported alongside it.

Part 5

Posterior Predictive Curves

Why move back to the original resistance scale?

The log-log model is fitted on the transformed response,

$$\log(Y_i) = x_i^\top \beta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

But the measured physical quantity is resistance itself, Y_i , not $\log(Y_i)$. For prediction we therefore simulate on the log scale and transform back:

$$\log(Y_i^{\text{rep}}) \sim \mathcal{N}(x_i^\top \beta, \sigma^2), \quad Y_i^{\text{rep}} = \exp\{\log(Y_i^{\text{rep}})\}.$$

Speaking point

Exponentiating is not cosmetic: it returns predictions to the resistance scale, keeps them positive, and makes the log-log predictive distribution lognormal.

What is Y^{rep} and why is μ_i exponential?

Posterior predictive checking generates replicated observations from the fitted model:

$$Y_i^{\text{rep}} \mid \theta^{(s)} \sim p(Y_i \mid \theta^{(s)}),$$

so each band includes both posterior parameter uncertainty and fresh observation noise.

For the heteroscedastic nonlinear model we predict directly on the resistance scale:

$$Y_i^{\text{rep}} \sim \text{Gamma}(\alpha_i, r_i), \quad \mu_i = \exp\{b_0 + \gamma_F \widetilde{\log Fn_i} + \beta_B \widetilde{bd_i}\}.$$

$$\sigma_i = \sigma_0 \mu_i^\delta, \quad \alpha_i = (\mu_i / \sigma_i)^2, \quad r_i = \mu_i / \sigma_i^2.$$

- The exponential form is the power law $kFn^{\gamma_F} \exp(\beta_B bd)$ written as a positive mean.
- $\delta = 0$ means constant variance; $\delta > 0$ lets uncertainty grow with resistance.

Which models are compared?

Variable selection came from BAS/BMA.

- Froude is forced into every model because it is the dominant physical predictor.
- Among the five hull covariates, `beam_draught` has the largest PIP (≈ 0.565).
- Evidence is only moderate, so the nonlinear stage treats it as a candidate, not a guaranteed effect.

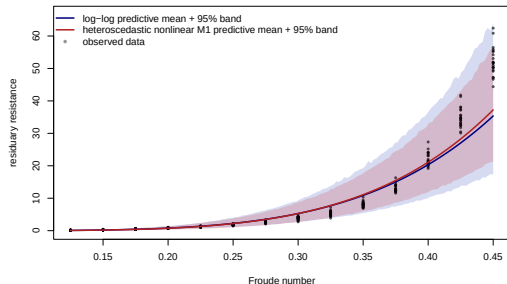
Model	Purpose in predictive comparison
Log-log	Baseline power law, back-transformed to Y
Gamma M1, $\delta = 0$	Same mean, constant variance
Gamma M1, δ learned	Same mean, heteroscedastic variance

Main question

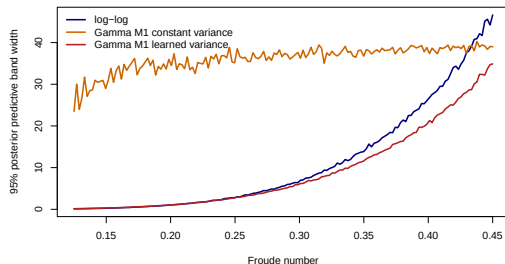
Does learning heteroscedasticity improve calibration without losing the Froude power-law interpretation?

Posterior predictive curves and band widths

Posterior predictive curves on the original scale



Predictive uncertainty across the Froude range



- Left: observed resistances are black points; bands are 95% posterior predictive intervals on the original scale.
- Log-log is positive after back-transformation and imposes multiplicative uncertainty.
- Gamma learned variance also stays positive, but estimates $\sigma_i = \sigma_0 \mu_i^\delta$ directly on the resistance scale.
- Right: constant variance is too narrow; learned variance widens intervals where resistance is larger.

Calibration and train/test check

Full-data posterior predictive calibration

Model	Cov.	Width
Log-log	0.968	12.241
Gamma const. var.	0.679	4.212
Gamma learned var.	0.932	5.403

Constant variance under-covers badly; learned variance restores near-nominal coverage with narrower bands than log-log.

80/20 train/test posterior predictive check

Model	Mean LPD	Cov.	Width
Log-log	-1.540	0.984	14.662
Gamma const.	-3.773	0.710	4.420
Gamma learned	-1.992	0.935	6.055

The split is a sanity check: log-log wins LPD but is widest; learned Gamma gives calibrated intervals on the physical scale.

Takeaway

The robust story is the Froude power law plus heteroscedastic uncertainty; the hull effect remains weak.